

Perturbed Coin Model Analysis

This notebook validates the symmetry-analysis infrastructure used in the perturbed coin model workflow.

It checks that the central classes and methods for tensor-product symmetry generators are available, including both the standard and $SO(N)$ -specific implementations.

It also verifies method signatures and wrapper consistency in `MatrixProductState`, ensuring coefficient extraction via `find_generators_coefficients` is correctly exposed for downstream analysis.

```
=====
MODULE VALIDATION CHECK
=====
```

- ```
[1] Checking Centraliser.TensorProductGroup...
 ✓ TensorProductGroup class exists with required methods

[2] Checking Centraliser.TensorProductGroupSO (SO(N) framework)...
 ✓ TensorProductGroupSO class exists with required methods

[3] Checking Centraliser.SON (SO(N) basis)...
 ✓ SON class exists with so_basis method

[4] Checking find_generators_coefficients parameters...
 ✓ Parameters: ['self', 'M', 'verbose', 'basis_generators', 'reference_coeffs']

[5] Checking MatrixProductState.find_generators_coefficients wrapper...
 ✓ Wrapper method exists with parameters: ['self', 'M', 'basis', 'verbose', 'dim', 'reference_coeffs']
```

## Form of the model

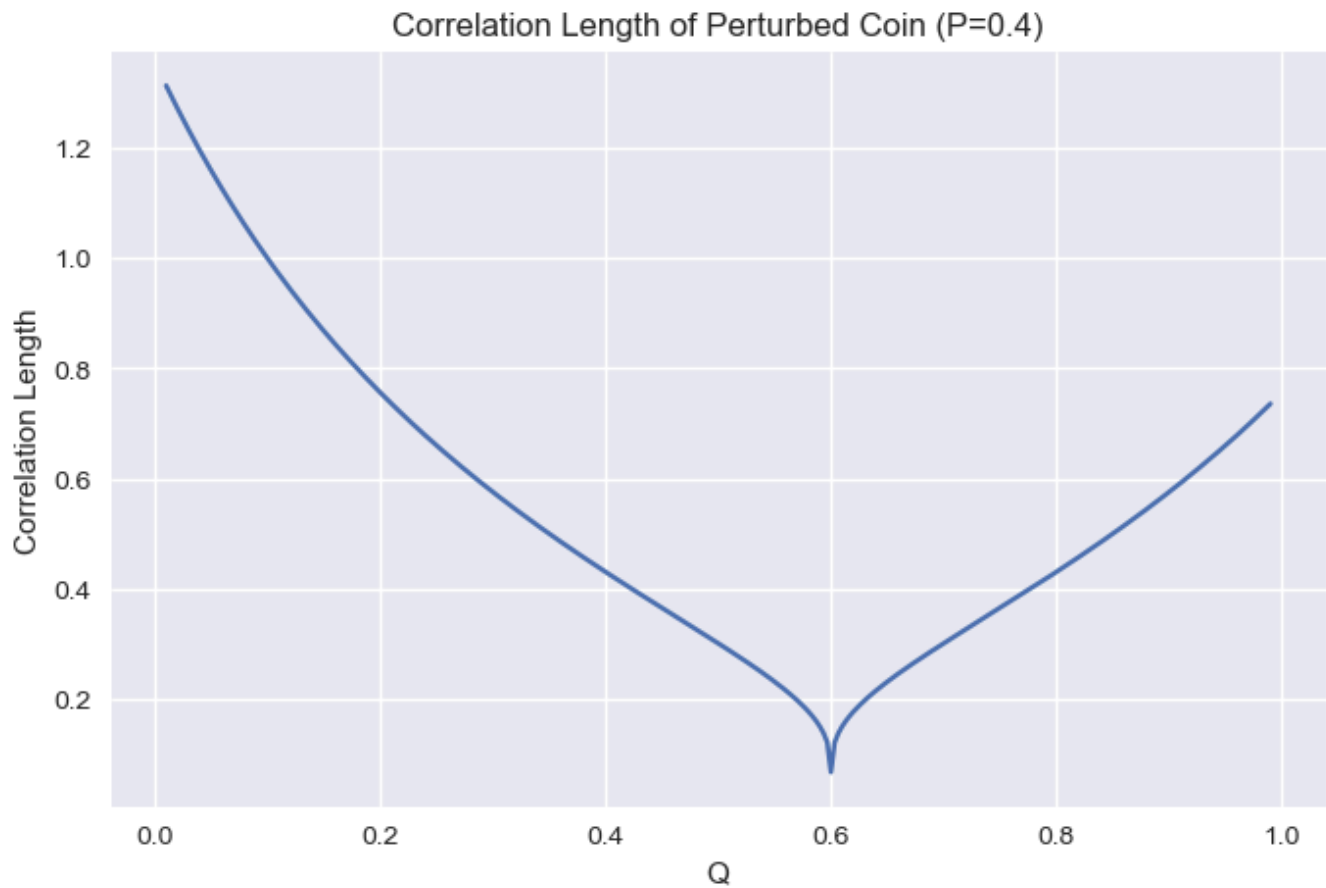
The site matrices are given by the form:

```
[[[0.6325 0.]
 [0.8944 0.]

 [[0. 0.7746]
 [0. 0.4472]]]
```

## Check Correlation Length

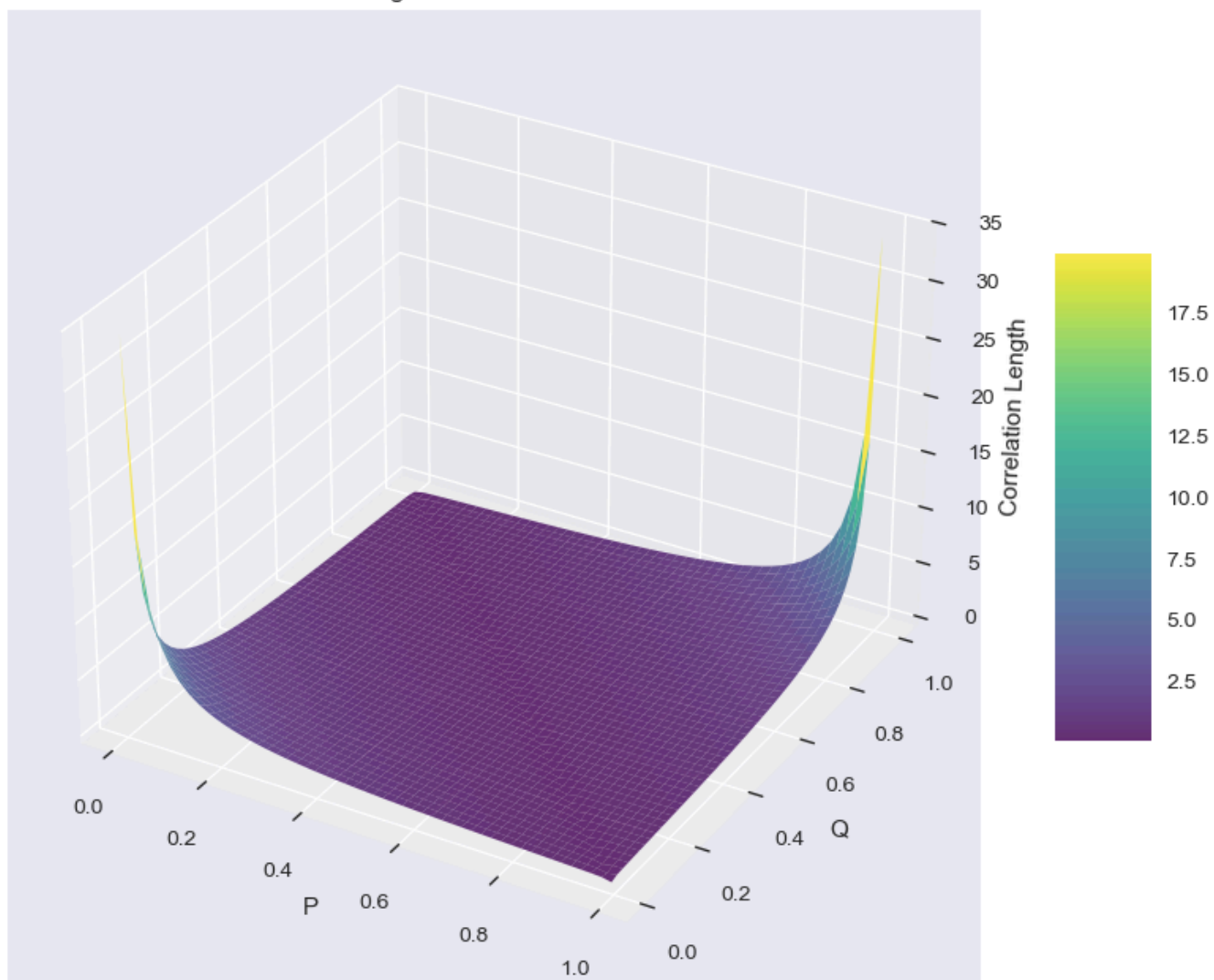
The system becomes memoryless when it approaches the line  $p = q$ , we can observe this with the plots below.



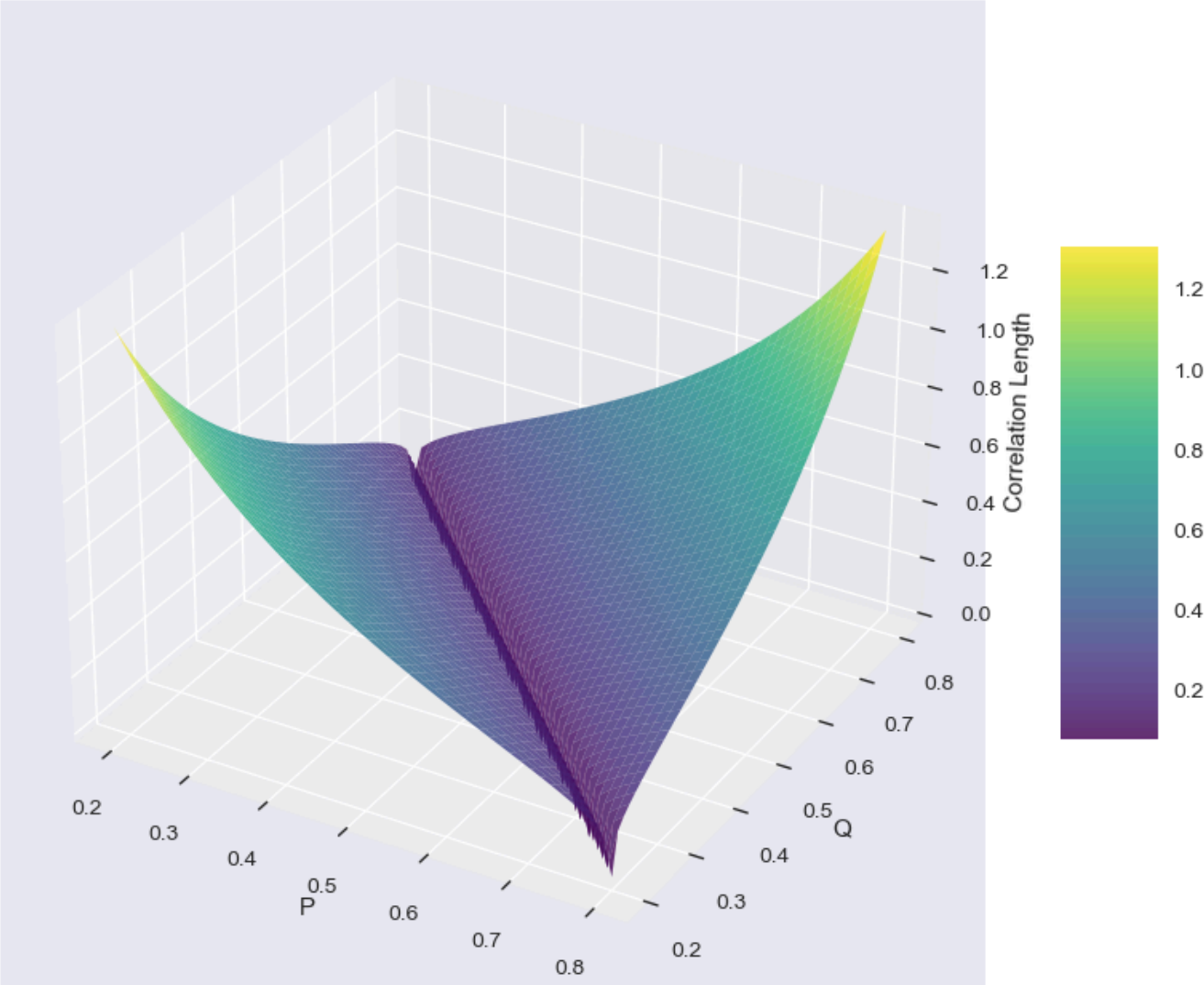
As we can see, the correlation length drops to 0 when  $p = 1 - q$ .

```
/Users/ernestchan/Desktop/EMachine/emachine_v2.py:1235: RuntimeWarning: divide by zero encountered in log2
 return -1 / np.log2(np.abs(second_largest))
```

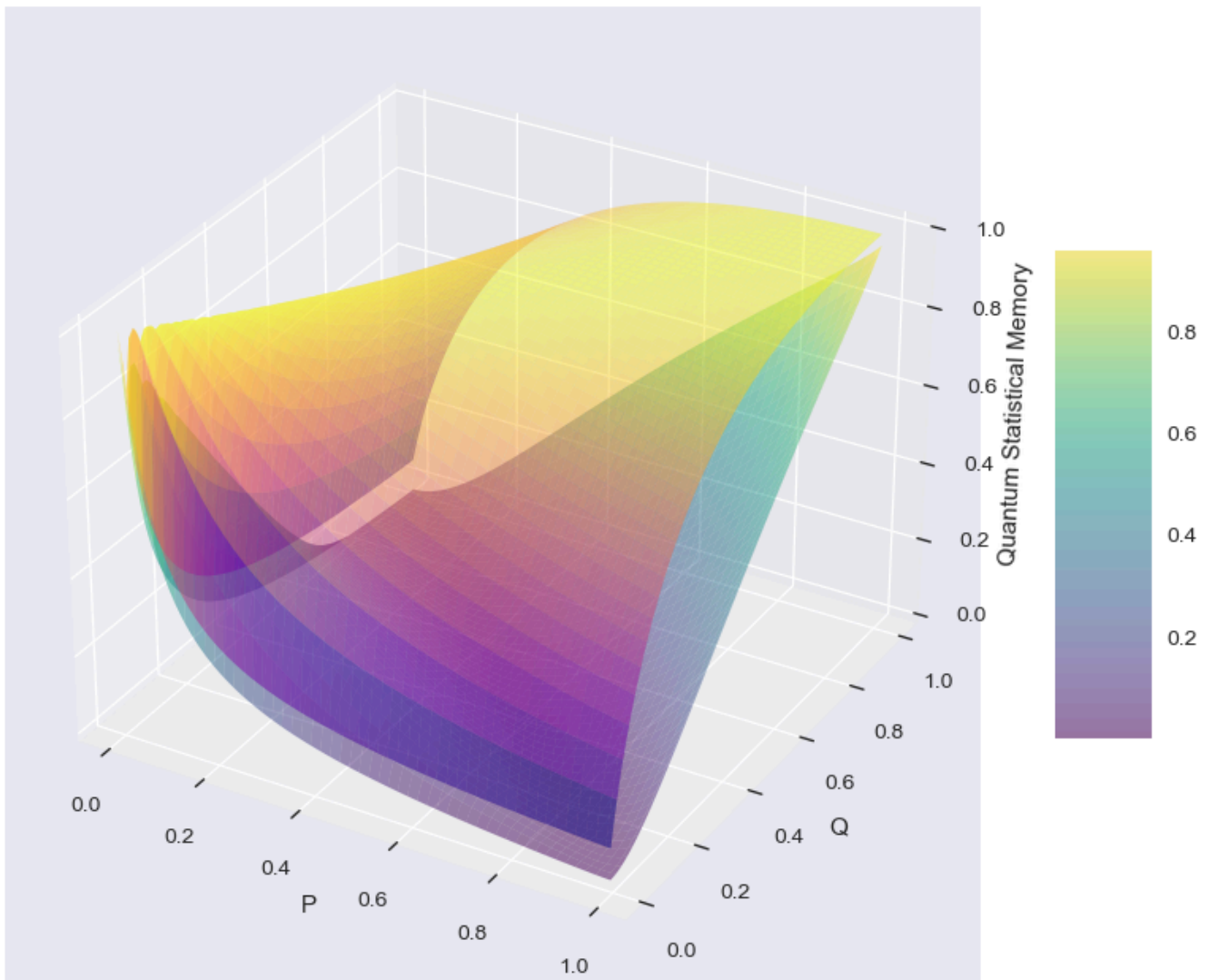
Correlation Length of Perturbed Coin vs P and Q



Correlation Length of Perturbed Coin vs P and Q



Quantum Statistical Memory of Perturbed Coin vs P and Q



## Checking the Produced Sequence by the Model

```
[0 0 1 0 1 1 0 0 0 0 0 1 0 1 1 1 1 1 1 0 0 0 1 0 0 0 1 1 0 0 1 1 1 0 1 1
 0 1 0]
```

## Symmetry Finding for the Perturbed Coin

We now attempt to find the symmetry group generators for the perturbed coin process for a fixed value of  $p \neq 1 - q$ .

The parent hamiltonian in consideration is:

```
[[0.0009 0.0058 -0.0029 -0.003 -0.0168 -0.003 0.0228 -0.0014]
 [0.0058 0.4543 -0.2884 -0.1988 0.1561 -0.1988 0.1533 -0.1943]
 [-0.0029 -0.2884 0.1877 0.1477 -0.1221 0.1477 -0.0761 0.0709]
 [-0.003 -0.1988 0.1477 0.7113 -0.0799 -0.2887 -0.0785 -0.2193]
 [-0.0168 0.1561 -0.1221 -0.0799 0.5042 -0.0799 -0.4429 -0.0384]
 [-0.003 -0.1988 0.1477 -0.2887 -0.0799 0.7113 -0.0785 -0.2193]
 [0.0228 0.1533 -0.0761 -0.0785 -0.4429 -0.0785 0.6015 -0.0377]
 [-0.0014 -0.1943 0.0709 -0.2193 -0.0384 -0.2193 -0.0377 0.8288]]
```

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#### SYMMETRIC GENERATORS ANALYSIS

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⚠ No symmetry found: The null space is empty.

=====

Now for the case  $p = q = 0.5$ ,

The parent hamiltonian in consideration is:

```
[[0.5 0. -0.5 0. 0. 0. 0. 0.]
 [0. 0.5 0. -0.5 0. 0. 0. 0.]
 [-0.5 0. 0.5 0. 0. 0. 0. 0.]
 [0. -0.5 0. 0.5 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.5 0. -0.5 0.]
 [0. 0. 0. 0. 0. 0.5 0. -0.5]
 [0. 0. 0. 0. -0.5 0. 0.5 0.]
 [0. 0. 0. 0. 0. -0.5 0. 0.5]]
```

Verifying that the MPS is in the ground space of the parent hamiltonian:

ph\_3\_3 @ state = [0. 0. 0. 0. 0. 0. 0. 0.]

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### SYMMETRIC GENERATORS ANALYSIS

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Found 1 generator(s) in the null space:

---

Generator 1:

Null space basis vector index: 0

Coefficients in local su(2) basis (orthonormalized): [1. 0. 0.]

Basis generators in su(2) (with non-zero coefficients):

[0] (coeff: 1.0):

[[0.+0.j 1.+0.j]

[1.+0.j 0.+0.j]]

Local su(2) generator (Hermitian enforced):

[[0.+0.j 1.+0.j]

[1.+0.j 0.+0.j]]

---

Orthonormalized null space basis (1 vectors):

Basis vector 1 (norm = 1.0000):

[1. 0. 0.]

---

We see that the hamiltonian is of the form  $H = I \otimes (I - \sigma_x) \otimes I$ .

Checking the form of the Hamiltonian:

```
[[0.5 0. -0.5 -0. 0. 0. -0. -0.]
 [0. 0.5 -0. -0.5 0. 0. -0. -0.]
 [-0.5 -0. 0.5 0. -0. -0. 0. 0.]
 [-0. -0.5 0. 0.5 -0. -0. 0. 0.]
 [0. 0. -0. -0. 0.5 0. -0.5 -0.]
 [0. 0. -0. -0. 0. 0.5 -0. -0.5]
 [-0. -0. 0. 0. -0.5 -0. 0.5 0.]
 [-0. -0. 0. 0. -0. -0.5 0. 0.5]]
```

# Finding Virtual Symmetry

Next we find the associated virtual symmetry of the system

Physical generator G:

```
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]
```

Virtual generator found:

Phase: 1.000000+0.000000j

Error: 2.160247e-06

Relative error: 1.527525e-06

Generator K:

```
[[1.+0.j -0.+0.j]
 [-0.+0.j -1.+0.j]]
```

The canonical form of the MPS is:

```
[[[0.7071 0.7071]
 [0. 0.]]
```

```
[[0.7071 -0.7071]
 [-0. 0.]]]
```

Verifying that the MPS is in the ground space of the parent hamiltonian:

```
[0. 0. 0. 0. 0. 0. 0. 0.]
```

Transfer Matrix:

```
[[1. 0. 0. 1.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
```

## Verifying the Generator Equation

Hence, we can conclude that rotations generated by  $\sigma_x$  on the physical level can be projected onto the virtual level as rotations generated by  $\sigma_z$ . As they are both 1-D subspaces of  $SU(2)$ , the 2-cocycle is trivially constant.



Generator 1

For angle  $\theta = 0.0 \pi$

Site matrix A[0]

| Physical                 | Virtual                  |
|--------------------------|--------------------------|
| -----                    | -----                    |
| [[0.7071+0.j 0.7071+0.j] | [[0.7071+0.j 0.7071+0.j] |
| [0. +0.j 0. +0.j]]       | [0. +0.j 0. +0.j]]       |

Site matrix A[1]

| Physical                   | Virtual                    |
|----------------------------|----------------------------|
| -----                      | -----                      |
| [[ 0.7071+0.j -0.7071+0.j] | [[ 0.7071+0.j -0.7071+0.j] |
| [-0. +0.j 0. +0.j]]        | [-0. +0.j 0. +0.j]]        |

Relative Phases: 0.0  $\pi$   
0.0  $\pi$   
0.0  $\pi$   
-0.0  $\pi$

For angle  $\theta = 0.25 \pi$

Site matrix A[0]

| Physical             | Virtual                     |
|----------------------|-----------------------------|
| -----                | -----                       |
| [[0.5+0.5j 0.5-0.5j] | [[ 0.7071+0.j -0. +0.7071j] |
| [0. -0.j 0. +0.j ]]  | [-0. -0.j 0. -0.j ]]        |

Site matrix A[1]

| Physical               | Virtual                     |
|------------------------|-----------------------------|
| -----                  | -----                       |
| [[ 0.5+0.5j -0.5+0.5j] | [[ 0.7071-0.j -0. -0.7071j] |
| [-0. +0.j 0. +0.j ]]   | [-0. +0.j 0. +0.j ]]        |

Relative Phases: 0.25  $\pi$   
-0.75  $\pi$   
0.25  $\pi$   
-0.75  $\pi$

For angle  $\theta = 0.3333 \pi$

Site matrix A[0]

| Physical | Virtual |
|----------|---------|
|----------|---------|

|                                  |  |                                |
|----------------------------------|--|--------------------------------|
| -----                            |  | -----                          |
| [[0.3536+0.6124j 0.3536-0.6124j] |  | [[ 0.7071+0.j -0.3536+0.6124j] |
| [0. -0.j 0. +0.j ]]              |  | [-0. -0.j 0. -0.j ]]           |

Site matrix A[1]

|                                    |  |                               |
|------------------------------------|--|-------------------------------|
| Physical                           |  | Virtual                       |
| -----                              |  | -----                         |
| [[ 0.3536+0.6124j -0.3536+0.6124j] |  | [[ 0.7071-0.j 0.3536-0.6124j] |
| [-0. +0.j 0. +0.j ]]               |  | [-0. +0.j 0. +0.j ]]          |

Relative Phases: 0.3333  $\pi$   
1.0  $\pi$   
0.3333  $\pi$   
-1.0  $\pi$

For angle  $\theta = 0.5 \pi$

Site matrix A[0]

|                          |  |                            |
|--------------------------|--|----------------------------|
| Physical                 |  | Virtual                    |
| -----                    |  | -----                      |
| [[0.+0.7071j 0.-0.7071j] |  | [[ 0.7071+0.j -0.7071+0.j] |
| [0.-0.j 0.+0.j ]]        |  | [-0. -0.j 0. -0.j]]        |

Site matrix A[1]

|                          |  |                          |
|--------------------------|--|--------------------------|
| Physical                 |  | Virtual                  |
| -----                    |  | -----                    |
| [[0.+0.7071j 0.+0.7071j] |  | [[0.7071-0.j 0.7071-0.j] |
| [0.+0.j 0.+0.j ]]        |  | [0. +0.j 0. +0.j]]       |

Relative Phases: 0.5  $\pi$   
0.5  $\pi$   
0.5  $\pi$   
0.5  $\pi$

For angle  $\theta = 0.75 \pi$

Site matrix A[0]

|                         |  |                             |
|-------------------------|--|-----------------------------|
| Physical                |  | Virtual                     |
| -----                   |  | -----                       |
| [[ -0.5+0.5j -0.5-0.5j] |  | [[ 0.7071-0.j -0. -0.7071j] |
| [-0. -0.j -0. +0.j ]]   |  | [-0. +0.j 0. +0.j ]]        |

Site matrix A[1]

|                        |  |                             |
|------------------------|--|-----------------------------|
| Physical               |  | Virtual                     |
| -----                  |  | -----                       |
| [[ -0.5+0.5j 0.5+0.5j] |  | [[ 0.7071+0.j -0. +0.7071j] |
| [ 0. +0.j -0. +0.j ]]  |  | [-0. -0.j 0. -0.j ]]        |

Relative Phases: 0.75  $\pi$   
-0.25  $\pi$   
0.75  $\pi$   
-0.25  $\pi$

=====

For angle  $\theta = 1.0 \pi$

Site matrix A[0]

| Physical                         |  | Virtual                        |
|----------------------------------|--|--------------------------------|
| -----                            |  | -----                          |
| [[ -0.7071+0.j -0.7071-0.j ]     |  | [[ 0.7071-0.j 0.7071-0.j ]     |
| [ -0.      -0.j -0.      +0.j ]] |  | [ 0.      +0.j 0.      +0.j ]] |

Site matrix A[1]

| Physical                        |  | Virtual                         |
|---------------------------------|--|---------------------------------|
| -----                           |  | -----                           |
| [[ -0.7071+0.j 0.7071+0.j ]     |  | [[ 0.7071+0.j -0.7071+0.j ]     |
| [ 0.      +0.j -0.      +0.j ]] |  | [ -0.      -0.j 0.      -0.j ]] |

Relative Phases: 1.0  $\pi$   
-1.0  $\pi$   
1.0  $\pi$   
-1.0  $\pi$

=====

# Discrete Symmetry of Parent Hamiltonian via Permutation Group

We investigate the permutation symmetry of symbols in a stochastic system by its Hamiltonian.

Permutation index 0 commutes with M.  
Permutation index 1 commutes with M.  
Total symmetries found: 2  
Symmetry indices: [0, 1]  
[array([[1.+0.j, 0.+0.j],  
[0.+0.j, 1.+0.j]]),  
array([[0.+0.j, 1.+0.j],  
[1.+0.j, 0.+0.j]])]

# Projection of Discrete Symmetry onto Virtual Level

The projection acts via the same equation,  $U^{xy}A_{ij}^y = e^{i\phi}V_{ik}A_{kl}^xV_{lj}^{-1}$ , but this time we don't have to Taylor expand the equation to get the equation for the generators. To solve this, we can do,  
 $U^{xy}A_{ij}^yV_{jk} - e^{i\phi}V_{ij}A_{jk}^x = 0 \Rightarrow B_{ij}^xV_{jk} - e^{i\phi}V_{ij}A_{jk}^x = 0$

```
[[[0. 0.70710678]
 [0. 0.70710749]]
```

```
[[0.70710678 0.]
 [0.70710607 0.]]]
[0. 0. 0. 0. 0. 0. 0. 0.]
```

We can conclude that in general, if  $p = q$ , for the dihedral symmetry group  $D_2 = \{e, d\}$  on the physical level, the element  $e$  corresponds to the set of gauge matrices,

$$e \rightarrow \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \quad (1)$$

, where  $a$  is a free elements. For the element  $d$  correesponds to matrices,

$$d \rightarrow \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix} \quad (2)$$

, where  $b$  is a free element.

Total solutions found: 2

Solution 1:

```
Permutation matrix:
[[1.+0.j 0.+0.j]
 [0.+0.j 1.+0.j]]
phi: 0.000000
Number of null vectors: 1
Null vector 1 (matrix form):
[[-0.7071+0.j 0. +0.j]
 [0. +0.j -0.7071+0.j]]
```

—

Solution 2:

```
Permutation matrix:
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]
phi: 0.000000
Number of null vectors: 1
Null vector 1 (matrix form):
[[0. +0.j 0.7071+0.j]
 [0.7071+0.j 0. +0.j]]
```

—